

CSCE 46103/56103 Artificial Intelligence – Fall 2026

Probability, Independence, and Bayes' Rule

Jungtaek Kim

`jungtaek.kim@uark.edu`

University of Arkansas

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Today's Topics

1. **Probability Foundations:** events, distributions, joint and conditional probabilities
2. **Independence and Bayes' Rule:** factorization, belief updates, and base rates
3. **Bayesian Networks:** directed graphs, local probabilities, and joint factorization
4. **Conditional Independence:** chains, forks, colliders, and graph reading

Opening Challenge: A Fault Alarm

A fictional machine sensor

- ▶ 1% of machines have a fault.
- ▶ 90% of faulty machines trigger an alarm.
- ▶ 5% of machines without a fault also trigger an alarm.

Prediction

An alarm has sounded. How likely is a fault?

About 15%, 50%, or 90%?

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The sensor's 90% detection rate answers a different question.

Variables, Values, and Events

- ▶ A random variable describes an uncertain quantity: F is fault status and A is alarm status.
- ▶ Lowercase letters denote events: f means $F = \text{true}$; a means $A = \text{true}$.
- ▶ $\neg f$ means no fault; (f, a) means both a fault and an alarm.

Notation	Meaning
$P(f)$	A single event's probability
$P(F)$	The distribution $\langle P(f), P(\neg f) \rangle$
$P(F, A)$	The joint distribution over both variables

Events and Probability Rules

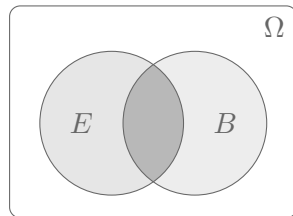
An event is a set of possible outcomes; Ω contains every possible outcome.

$$0 \leq P(E) \leq 1,$$

$$P(\Omega) = 1,$$

$$P(\neg E) = 1 - P(E),$$

$$P(E \cup B) = P(E) + P(B) - P(E \cap B).$$



Shaded overlap: $E \cap B$

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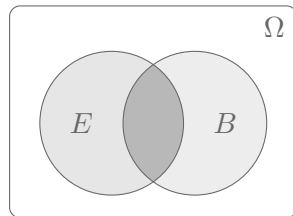
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Shaded overlap: $E \cap B$

Adding $P(E)$ and $P(B)$ counts their overlap twice. Subtract it once.

Discrete Probability Distributions

For a discrete variable X with possible values $x_1, \dots, x_k$,

$$P(X) = \langle P(x_1), \dots, P(x_k) \rangle, \quad \sum_{i=1}^k P(x_i) = 1.$$

Fault status

$$P(F) = \langle 0.01, 0.99 \rangle.$$

The order is f , then $\neg f$.

A conditional distribution also sums to one for each fixed condition.

Sensor response

$$P(A \mid f) = \langle 0.90, 0.10 \rangle.$$

The order is a , then $\neg a$; a fault is given.

The Joint Distribution

A complete table assigns a probability to every combination of values.

	Alarm a	No alarm $\neg a$
Fault f	0.0090	0.0010
No fault $\neg f$	0.0495	0.9405

$$P(f, a) = 0.0090, \quad \sum_{f' \in \{f, \neg f\}} \sum_{a' \in \{a, \neg a\}} P(f', a') = 1.$$

Each machine belongs to exactly one cell. We will derive the cells from the sensor model.

Marginalization: Summing Out a Variable

To find a probability without observing A , add over its possible values.

$$P(f) = P(f, a) + P(f, \neg a) = 0.009 + 0.001 = 0.01.$$

	a	$\neg a$	$P(F)$
f	0.0090	0.0010	0.01
$\neg f$	0.0495	0.9405	0.99
$P(A)$	0.0585	0.9415	1

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In general, $P(x) = \sum_y P(x, y)$: retain X , sum out Y .

Conditioning: Restricting and Renormalizing

Given a fault, restrict attention to the fault row of the joint table.

$$P(a \mid f) = \frac{P(a, f)}{P(f)} = \frac{0.009}{0.01} = 0.90.$$

	Alarm a	No alarm $\neg a$
Joint mass with f	0.009	0.001
Divide by $P(f) = 0.01$	0.90	0.10

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Divide by $P(f) = 0.01$	0.90	0.10

In general, $P(E \mid B) = P(E \cap B)/P(B)$ when $P(B) > 0$.

The Product Rule

Rearrange the definition of conditional probability:

$$P(f, a) = P(a \mid f)P(f) = P(f \mid a)P(a).$$

Fault and alarm

$$P(f, a) = 0.90 \times 0.01 = 0.009.$$

No fault and alarm

$$P(\neg f, a) = 0.05 \times 0.99 = 0.0495.$$

The Product Rule

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No fault and alarm

$$P(\neg f, a) = 0.05 \times 0.99 = 0.0495.$$

Both directions describe the same joint event. Neither direction requires independence.

The Chain Rule

Repeated application of the product rule gives

$$P(x, y, z) = P(x)P(y \mid x)P(z \mid x, y).$$

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$$P(x, y, z) = P(x)P(y \mid x)P(z \mid x, y).$$

For any ordering of discrete variables,

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid x_1, \dots, x_{i-1}).$$

- ▶ The first factor is $P(x_1)$, with no conditioning variables.
- ▶ This identity does not assume independence.
- ▶ Independence can be assumed to simplify the conditioning sets.

Independence and Factorization

X and Y are independent, written $X \perp\!\!\!\perp Y$, if

$$P(x, y) = P(x)P(y) \quad \text{for every } x, y.$$

Equivalently, $P(x \mid y) = P(x)$ whenever $P(y) > 0$.

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The fault and alarm are dependent

$$P(a | f) = 0.90 \neq 0.0585 = P(a).$$

Knowing there is a fault changes the probability of an alarm.

Independence is an assumption about a distribution, not a default rule.

Independence and Mutually Exclusive Events

Independent events

$$P(E \cap B) = P(E)P(B).$$

Learning that one occurred leaves the other's probability unchanged.

Mutually exclusive events

$$P(E \cap B) = 0.$$

They cannot occur together: $E \cap B = \emptyset$.

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Independent events

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Learning that one occurred leaves the other's probability unchanged.

If $P(E) > 0$ and $P(B) > 0$, mutually exclusive events are dependent.

$$P(E \mid B) = 0 \neq P(E).$$

For example, f and $\neg f$ cannot both hold for one machine.

Mutually exclusive events

$$P(E \cap B) = 0.$$

They cannot occur together: $E \cap B = \emptyset$.

Checkpoint: Independent Coin Tosses

Assume two independent fair coin tosses. Define three events:

$$E = \{HH, HT\}, \quad B = \{HH, TH\}, \quad C = \{TH, TT\}.$$

E : first toss heads; B : second toss heads; C : first toss tails.

1. Are E and B independent? Can they occur together?
2. Are E and C independent? Can they occur together?

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E : first toss heads; B : second toss heads; C : first toss tails.

1. Are E and B independent? Can they occur together?
2. Are E and C independent? Can they occur together?

$$P(E \cap B) = \frac{1}{4} = P(E)P(B)$$

independent; both occur in HH,

$$P(E \cap C) = 0 \neq \frac{1}{4}$$

dependent; mutually exclusive.

Bayes' Rule from the Product Rule

The same joint probability has two factorizations:

$$P(f \mid a)P(a) = P(a \mid f)P(f).$$

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Dividing by $P(a) > 0$ gives

$$\underbrace{P(f \mid a)}_{\text{posterior}} = \frac{\overbrace{P(a \mid f)}^{\text{likelihood}} \overbrace{P(f)}^{\text{prior}}}{\underbrace{P(a)}_{\text{evidence probability}}}.$$

- ▶ The prior describes belief before the alarm is observed.
- ▶ The likelihood describes how probable the observed alarm is under a fault.
- ▶ The posterior describes belief after the alarm is observed.

Total Probability for the Evidence

An alarm can occur with a fault or without a fault. These cases partition the possibilities.

$$\begin{aligned}P(a) &= P(a, f) + P(a, \neg f) \\&= P(a \mid f)P(f) + P(a \mid \neg f)P(\neg f) \\&= 0.90(0.01) + 0.05(0.99) = 0.0585.\end{aligned}$$

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For mutually exclusive, exhaustive hypotheses $h_1, \dots, h_k$,

$$P(a) = \sum_{i=1}^k P(a \mid h_i)P(h_i).$$

Opening Challenge: Posterior Fault Probability

$$P(f \mid a) = \frac{0.90(0.01)}{0.90(0.01) + 0.05(0.99)} = \frac{0.009}{0.0585} = \frac{2}{13} \approx 15.38\%.$$

Opening Challenge: Posterior Fault Probability

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Detection rate

$P(a | f) = 90\%$: among faulty machines, how many trigger alarms?

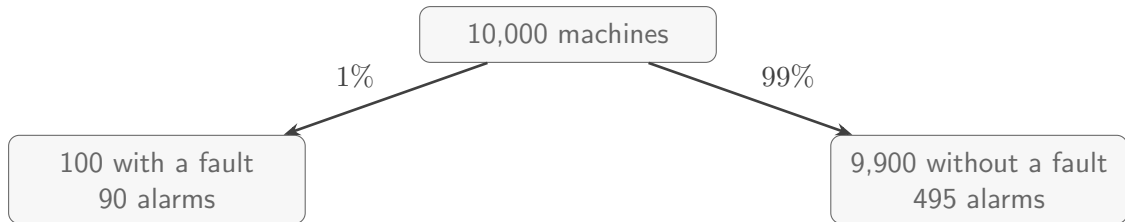
Posterior probability

$P(f | a) \approx 15.38\%$: among machines with alarms, how many have faults?

An alarm raises the fault probability from 1% to about 15%; a fault remains less likely than no fault.

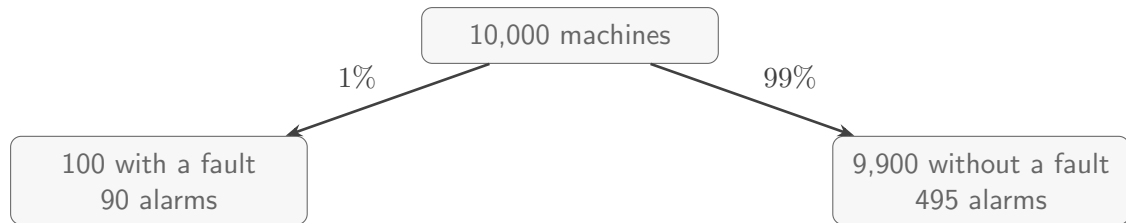
The Same Update in Counts

Consider an illustrative population of 10,000 machines matching the assumed rates.



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$$P(f | a) = \frac{\text{faulty machines with alarms}}{\text{all machines with alarms}} = \frac{90}{90 + 495} = \frac{2}{13}.$$

The large no-fault population contributes most of the alarms.

Practice: A Different Base Rate

In a different operating environment, assume $P(f) = 0.10$. The same sensor has $P(a \mid f) = 0.90$ and $P(a \mid \neg f) = 0.05$.

1. Predict whether $P(f \mid a)$ increases or decreases.
2. Compute the new posterior probability.

Practice: A Different Base Rate

In a different operating environment, assume $P(f) = 0.10$. The same sensor has $P(a | f) = 0.90$ and $P(a | \neg f) = 0.05$.

1. Predict whether $P(f | a)$ increases or decreases.
2. Compute the new posterior probability.

$$P(f | a) = \frac{0.90(0.10)}{0.90(0.10) + 0.05(0.90)} = \frac{0.09}{0.135} = \frac{2}{3} \approx 66.67\%.$$

The likelihoods stayed fixed. The changed prior changed the posterior.

Posterior Distributions and Normalization

Return to the original prior $P(f) = 0.01$. Update both possible fault states:

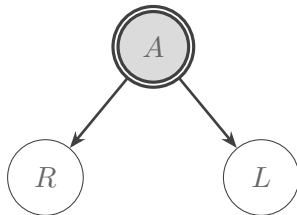
$$\begin{aligned}P(F | a) &= \alpha \langle P(a | f)P(f), P(a | \neg f)P(\neg f) \rangle \\&= \alpha \langle 0.009, 0.0495 \rangle, \quad \alpha = \frac{1}{0.009 + 0.0495} \\&= \left\langle \frac{2}{13}, \frac{11}{13} \right\rangle.\end{aligned}$$

- ▶ Multiply the prior of each state by its evidence likelihood.
- ▶ Normalize the resulting weights so that they sum to one.

Conditional Independence

$X \perp\!\!\!\perp Y \mid Z$ means that for every z with $P(z) > 0$,

$$P(x, y \mid z) = P(x \mid z)P(y \mid z) \quad \text{for every } x, y.$$



Double border: observed

A modeling assumption

R : remote notification; L : warning light. Once alarm status A is known, assume that R gives no additional information about L .

$$R \perp\!\!\!\perp L \mid A.$$

Conditional and Marginal Independence

The assumption $R \perp\!\!\!\perp L \mid A$ does not imply $R \perp\!\!\!\perp L$.

Alarm status unknown

A notification may suggest an alarm. An alarm may make the warning light more likely.

$P(l \mid r)$ may differ from $P(l)$.

Alarm status known

Under the assumed model, learning about the notification adds no information about the light.

$$P(l \mid r, a) = P(l \mid a).$$

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The assumption $R \perp\!\!\!\perp L \mid A$ does not imply $R \perp\!\!\!\perp L$.

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Which variables we condition on is part of every independence claim.

Probability Takeaways

- ▶ Joint distributions describe complete assignments; marginalization sums out variables.
- ▶ Conditioning restricts the possibilities and renormalizes their probabilities.
- ▶ Independence permits factorization and must be justified by the model.
- ▶ Bayes' rule combines the prior with the evidence likelihood to obtain a posterior.

References

S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL <https://aima.cs.berkeley.edu/>.